Beyond Differential Expression:

Causal Effect Estimation in CRISPR Perturb-seq with IV, SV, and Two-Step GMM

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1 Introduction

Background

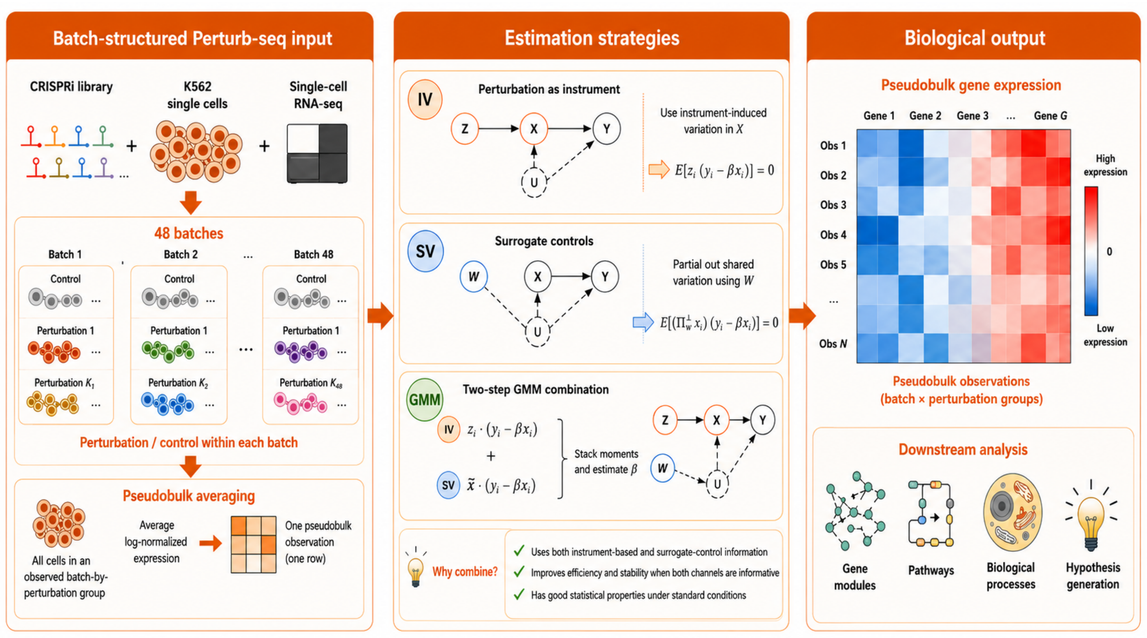
Perturb-seq jointly measures CRISPR perturbations and transcriptome-wide responses, enabling large-scale maps of gene function [4]. Standard differential-expression analysis identifies genes that change between perturbed and control samples, but it does not directly estimate how variation in target-gene expression affects downstream genes.

Challenge

Target-gene expression and downstream-gene expression may both depend on latent cell state, sequencing depth, cell-cycle effects, and batch variation. Therefore, a direct regression of downstream expression on target expression may be confounded.

Research objective

Using K562 CRISPRi Perturb-seq data, we study how reducing the expression of a target gene changes the expression of downstream genes. Cells carrying the same perturbation within each batch are summarized as batch-level pseudobulk observations. We combine two complementary sources of deconfounding information:
(i) the perturbation indicator as an **instrumental variable**, and
(ii) expression factors from common non-DE genes as **surrogate controls**.
We compare IV, SV, and optimally weighted **two-step GMM** estimators to identify robust downstream responses and clarify gene-regulatory relationships.



2 Methodology

For each target, each downstream gene is analysed separately with $y_i = x_i^\top \beta + \varsigma_i$, where x_i is target-gene expression, y_i is downstream-gene expression, and β is the target effect. The error ς_i absorbs unobserved cell-state and technical variation, so direct regression may be confounded. The perturbation z_i provides an instrument; w_i (first 10 PCs from 1,000 non-DE genes) provides surrogate controls.

IV — Instrumental Variables

Uses only perturbation-induced variation in x_i via $\mathbb{E}(z_i\varsigma_i) = 0$:

$$\widehat{\beta}_{\text{iv}} = \left(X^\top P_Z X\right)^{-1} X^\top P_Z Y.$$

SV — Surrogate Controls

Partials out the component of x_i explained by w_i, $\Pi_w^{\perp} x_i = x_i - \Sigma_{xw}\Sigma_w^{-1} w_i$, and uses the residual variation:

$$\widehat{\beta}_{\text{sv}} = \left(X^\top P_W^{\perp} X\right)^{-1} X^\top P_W^{\perp} Y.$$

Optimally Weighted Two-Step GMM

Stack both moment conditions with $h_i = \left(z_i,\ \Pi_w^{\perp} x_i\right)^\top$ so that $\mathbb{E}[h_i(y_i - x_i^\top \beta)] = 0$. Define

$$\bar{g}(\theta) = \frac{1}{n}\begin{pmatrix} Z^\top (Y - X\theta) \\ X^\top P_W^{\perp} (Y - X\theta) \end{pmatrix}.$$

In the first step, use $M^{(0)} = I$ to obtain $\widehat{\beta}^{(1)} = \arg\min_\theta \bar{g}(\theta)^\top \bar{g}(\theta)$. Using its residuals, estimate the batch-clustered moment covariance $\widehat{\Omega}$. In the second step, use the optimal weight $M^\star = \widehat{\Omega}^{-1}$ [2]:

$$\widehat{\beta}_{\text{gmm}} = \arg\min_\theta\ \bar{g}(\theta)^\top \widehat{\Omega}^{-1} \bar{g}(\theta).$$

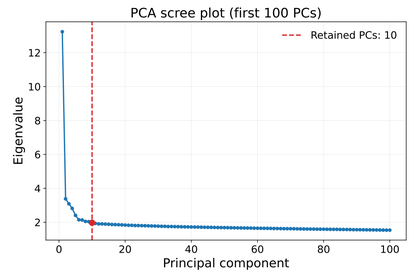
Asymptotic inference and efficiency

Under valid IV and SV moment conditions, sufficient identifying variation, finite score variances, and full-rank identification, all three estimators are consistent and asymptotically normal: $\sqrt{n}\left(\widehat{\beta}_m - \beta\right) \xrightarrow{d} \mathcal{N}(0, V_m),\ m \in \{\text{iv}, \text{sv}, \text{gmm}\}$. For the combined estimator, the optimally weighted GMM variance is $V^\star_{\text{comb}} = \left(J^\top \Omega^{-1} J\right)^{-1}$. When both moment conditions are valid, $V^\star_{\text{comb}} \preceq V^\star_{\text{iv}},\ V^\star_{\text{comb}} \preceq V^\star_{\text{sv}}$.

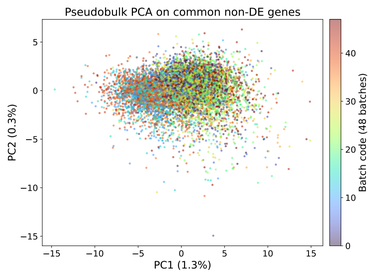
3 Results

1. Data and variable construction

We analysed **426,000** perturbation–outcome pairs (**213 perturbations** × **2,000 candidate downstream genes**) using batch-level pseudobulk observations. For example, for *TTK–SPC25*, X is *TTK* expression, Y is *SPC25* expression, Z indicates *TTK* perturbation versus control, and W contains the first **10 PCs** of **1,000 common non-DE genes** as surrogate controls. Significance was assessed using within-perturbation BH-adjusted q-values at $q \le 0.05$.



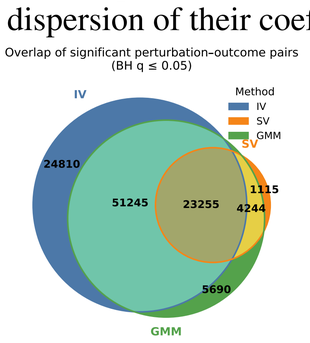
Choosing the dimension. The scree curve flattens near PC10, so we retain the **first 10 PCs**.



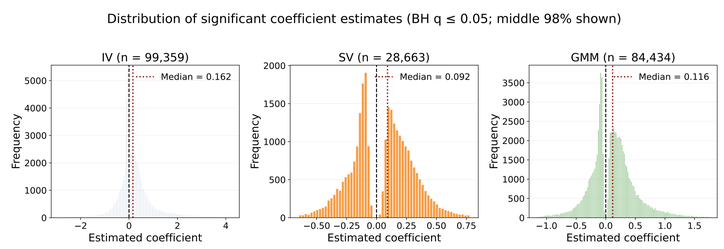
PCA scores by batch. PC1–PC2 scores of pooled pseudobulk observations, coloured by the **48 batches**.

2. IV, SV, and GMM estimates

We compare the three estimators by the overlap of their significant discoveries and the dispersion of their coefficient estimates.



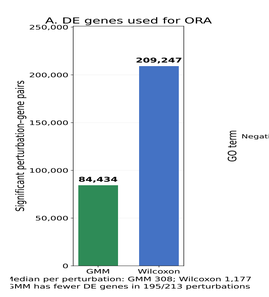
Overlap of significant pairs. IV/SV/GMM yielded **99,310 / 28,614 / 84,434** hits; **23,255** were shared by all three. GMM recovered **96.1%** of SV and **75.0%** of IV hits.

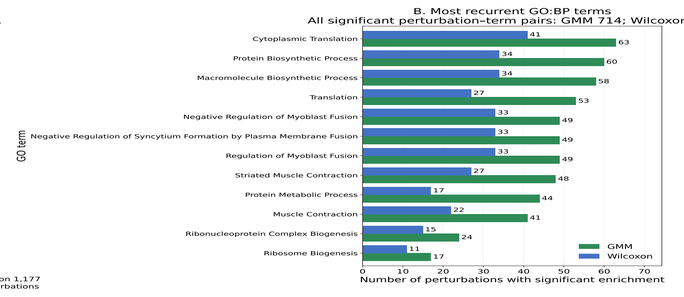


Distribution of significant estimates. All three distributions peak near zero and are right-skewed. IV is the widest, SV the narrowest, and GMM intermediate, with medians **0.162 / 0.092 / 0.116**.

3. Gene Ontology biological-process enrichment

For each perturbation, we apply over-representation analysis (ORA) to differentially expressed (DE) genes selected by GMM and Wilcoxon, and record significant Gene Ontology Biological Process (GO:BP) terms.
GMM selected fewer DE perturbation–gene pairs than Wilcoxon (**84,434 vs. 209,247**) but yielded more significant perturbation–GO associations (**714 vs. 436**). Translation and protein-biosynthesis terms also recurred across more perturbations, suggesting greater functional concentration and biological coherence.





Key findings. IV is most sensitive and dispersed, SV most conservative, and GMM intermediate. Despite fewer DE pairs than Wilcoxon, GMM yields more significant perturbation–GO associations (**714 vs. 436**), indicating greater biological coherence.

4 Conclusions

- **IV and SV give contrasting results.** IV yields the most discoveries and widest distribution; SV yields the fewest and narrowest.
- **GMM lies between the two.** It recovers **75.0%** of IV and **96.1%** of SV hits, with intermediate dispersion.
- **GMM shows stronger biological coherence.** Despite fewer DE pairs than Wilcoxon (**84,434 vs. 209,247**), it yields more significant perturbation–GO associations (**714 vs. 436**).

5 Future Work

- **Theory and simulation.** Extend the identification, consistency, and efficiency results to biased and locally misspecified moments, and quantify finite-sample bias, type-I error, power, and coverage across sample size, instrument strength, surrogate quality, and confounding level.
- **Real-data analysis.** Assess sensitivity to the number of PCs, gene selection, and batch processing, and extend to more genes and gene-regulatory pathways.

References

[1] Anderson, T. W. and Rubin, H. (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *Ann. Math. Statist.*, 20(1):46–63.
[2] Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4):1029–1054.
[3] Miao, W., Geng, Z. and Tchetgen Tchetgen, E. J. (2018). Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105(4):987–993.
[4] Replogle, J. M. et al. (2022). Mapping information-rich genotype–phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575.